Representation Theory of Finite Groups

Supplementary Midterm Examination

October 10, 2024

Instructions: All questions carry ten marks. Vector spaces and representations are assumed to be finite dimensional over the field of complex numbers.

1. Let R be a ring and M and N be two R-modules. Prove that the tensor product $M \otimes_R N$ is isomorphic to $N \otimes_R M$ as an R-module.
2. Let G be a group of odd order. Let $x \in G$ be a non-identity element. Prove that x^{-1} does not belong to the conjugacy class of x.
3. Let V be an irreducible representation of a finite group G. Determine all G-equivariant linear operators of V.
4. Let G be a non-abelian group of order 55. Determine all one-dimensional representations of G.
5. Let H be a subgroup of a finite group G. Prove that the regular representation of G is induced from the regular representation H.